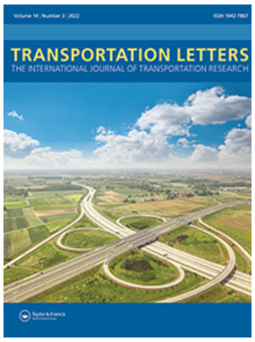

Taylor & Francis
Taylor & Francis Group

Transportation Letters
The International Journal of Transportation Research

ISSN: (Print) (Online) Journal homepage: www.tandfonline.com/journals/ytrl20

Multi-objective route planning problem for cycle-tourists

Siying Zhu

To cite this article: Siying Zhu (2022) Multi-objective route planning problem for cycle-tourists, Transportation Letters, 14:3, 298-306, DOI: 10.1080/19427867.2020.1860355

To link to this article: https://doi.org/10.1080/19427867.2020.1860355

Published online: 23 Dec 2020.

TRANSPORTATION LETTERS
2022, VOL. 14, NO. 3, 298–306
https://doi.org/10.1080/19427867.2020.1860355

Multi-objective route planning problem for cycle-tourists

Siying Zhu

School of Civil and Environmental Engineering, Nanyang Technological University, Singapore

ABSTRACT
Cycle-tourism which is a type of niche and sustainable tourism is increasingly promoted by transport and tourism agencies in recent years, but there are still few researches on the cycle-tourist route planning. This paper aims to formulate the cycle-tourist route-planning problem, in order to maximize cycle-tourists' utility of visiting points of interest (POIs), minimize the total travel time, maximize the Bicycle Level of Service (BLOS) and minimize the number of intersections on the cycle route taken, subject to monetary and time budget, etc. A multi-objective mixed integer linear programming model is formulated to optimize the cycle-tourist route plan. The problem is solved by augmented $\in$-constraint method (AUGMECON) whereby a set of nondominated solutions are generated to address the trade-offs among multiple objectives. The proposed model can help local government and agencies promote cycle-tourism and help cycle-tourists better plan the route to satisfy their personal requirements in a sustainable transportation system.

ARTICLE HISTORY
Received 04 December 2019
Revised 12 June 2020
Accepted 11 November 2020

KEYWORDS
Cycle-tourism; Route planning; Multi-objective mixed integer linear programming; Augmented - constraint method; Sustainable transportation system

Introduction

In recent years, tourism-related transport is increasingly considered as a major cause of environmental degradation (Chen and Cheng 2016). Moreover, environmental sustainability is closely related to the transportation sector, as emissions and noise caused by automobiles have a negative influence on the environment and human health (Szeto, Jaber, and Wong 2012; Szeto 2016; Majumdar, Mitra, and Pareekh 2019). Cycle-tourism, as an important mode of holiday transport, for which cycling is used as the primary mode of transportation, has drawn the attention of transport and tourism agencies to meet the principles of sustainable development at a destination level (Lumsdon 2000). In Europe, tourism-related cycle route-construction projects have been well planned and implemented (Ritchie 1998) and the cycle tourism has also been well developed in both rural and nonrural regions (Gazzola et al. 2018). With the development of cycle-tourism, environment, social, and economic benefits can be triggered in the tourist areas and the wider community (Lumsdon and McGrath 2011). Instead of standard tours, there is a trend that tourists increasingly prefer personalized options of tours (Hyde and Lawson 2003). The tour planning process is complex as tourists need to gather information from different sources and select a number of attraction points or points of interest (POIs) to visit according to their personal interests. Additionally, it is also essential to incorporate the realistic route choice factors with recent studies on cyclists' behavior (Hrnčř et al. 2017). Different from tourists traveling with automobiles who care most about the utility of visiting POIs and the travel time, the cycling comfort and safety level on the cycling route taken is also significant to cycle-tourists.

Although cycle-tourism which is a type of niche and sustainable tourism is increasingly promoted in recent years, the research gap remains as there are few researches on the cycle-tourist route planning. To promote cycle-tourism and meet the requirement of cycle-tourists, in this paper, we would like to formulate a multi-objective cycle-tourist route-planning problem, which considers cycle-tourist's utility of visiting POIs, the travel time, the Bicycle Level of Service (BLOS) of the trip, and the number of intersections on the cycling route, subject to constraints like monetary and time budget. No previous research has applied a multi-objective optimization model to cater to the various needs of cycle-tourists, to the authors' best knowledge. The result of this paper can help plan the tour for cycle-tourists and serve as the basis for the development of a mobile application for cycle-tourist route planning. The paper may further provide insights for tourism-related bike-way network design.

Related work

This paper addressed the cycle-tourist route-planning problem with a multi-objective optimization model, in consideration of the cycle-tourist's utility of visiting POIs, the travel time, the Bicycle Level of Service (BLOS) of the trip, and the number of intersections on the cycling route, subject to constraints like monetary and time budget. The following literature review contains four sections, namely, tourist route-planning problem, cyclist route choice problem and cycle-tourism, and discussion in order to provide an overview of the idea of the cycle-tourist route-planning problem.

Tourist route-planning problem

Hyde and Lawson (2003) suggested that tourists increasingly prefer more personalized trips to standard tours. The tourist trip design problem (TTDP) or tourist route-planning problem, which has attracted much attention recently, was firstly introduced by Vansteenwegen and Van Oudheusden (2007). TTDP refers to a tourist's route-planning problem for visiting multiple attractions or points of interest (POIs) (Gavalas et al. 2014). A profit value that can be seen as the utility of visiting certain POI in comparison with others is associated with each POI. The main objective of TTDP is to maximize the collected profit or the number of visited points subject to constraints, including entrance fees, distance, daily available time for sightseeing, etc. (Gunawan, Lau, and Vansteenwegen 2016). The profit value or attractiveness score of each POI should be determined first in order to solve the TTDP. According to the

CONTACT Siying Zhu SIYING001@e.ntu.edu.sg Nanyang Technological University , 639798 Singapore

interest of each tourist, information retrieval, or data mining techniques are applied to calculate the profit value for each POI (Souffriau et al. 2008; Gavalas et al. 2015). Basically, there are two types of TTDP variants, namely, single-tour TTDP and multiple-tour TTDP. Single-tour TTDP aims to find the single tour that maximizes the collected profit, for example, Zheng, Liao, and Qin (2017) designed personalized day tour route within tourist attractions with a four-step heuristic algorithm; Abbaspour and Samadzadegan (2011) focused on time-dependent tour planning in urban areas and chronological sequences of POIs over a specific period of time via various transportation modes were determined. On the other hand, multiple-tour TTDP aims to find multiple tours within a few days of the tourist's visit, for example, Gavalas et al. (2015) designed a context-aware web/mobile application eCOMPASS, generating personalized multi-modal tours for selected attraction points; Kotiloglu et al. (2017) proposed personalized multi-period tour recommendations.

Cyclist route choice problem

People will use bicycles for more trips if they get trip information in advance (Su et al. 2010; Hochmair 2005). Different from automobile drivers who care about the travel time and distance most, a significantly broader range of factors are considered by cyclists when selecting their cycle routes, including slope, traffic volumes, etc. (Broach, Dill, and Gliebe 2012). The route choice decisions of cyclists have been studied in many literatures. Stated preference (SP) data have been commonly used in literature to evaluate bicycle route choice from hypothetical choice situations, nevertheless, a major limitation exists due to the difference between claimed and observed behavior (Sener, Eluru, and Bhat 2009; Winters et al. 2011). Moreover, there are also revealed preference (RP) studies that estimated cyclist route choice model based on GPS observations of actual commuting routes (Zimmermann, Mai, and Frejinger 2017; Broach, Dill, and Gliebe 2012; Hood, Sall, and Charlton 2011).

It can be a complex task to find routes that consider all the factors affecting cyclists' route decisions, especially when the cycling environment is complicated. Therefore, intelligent route-planning software that can help find routes that best suit the needs and preferences of cyclists, especially those who are unfamiliar with their surroundings is necessary. Hochmair and Fu (2009) developed a web-based trip planner for cyclists, which is able to generate the fastest, safest, simplest, most scenic, and a shortest route, respectively. Su et al. (2010) proposed a web-based cycling route planner to generate a route based on one of the user-stated preference objectives, regarding distance, traffic pollution, etc. Turverey et al. (2010) designed a web-based tool for cyclist's route planner, which applied a weighted-sum method to address the trade-off between route safety and distance. However, if different or more cyclists participate in the survey, the resulting weight combination may be significantly different. Although these studies consider various objectives for the cyclist routing problem, the multi-criteria search methods were not applied to solve the multi-objective optimization problem. Furthermore, a multi-criteria bicycle routing problem was formulated by Song et al. (2014) and a full set of Pareto routes were found with an optimum multi-label correcting algorithm. Hrnčř et al. (2017) was build on Song et al. (2014), to formulate a multi-criteria bicycle routing problem to satisfy various requirements of cyclists and further proposed several speed-up heuristics. However, these two papers only focused on the derivation of route choices based on pre-determined start and end points. The selection of points of interest, which is significant for cycle-tourists, was not taken into consideration.

Cycle-tourism

Recent evidence suggested that cycle-tourism is regaining its popularity as a worldwide leisure and sustainable tourism activity, despite that cycle-tourism had once declined as a result of the growth of motorized traffic (Lee 2014). Cycle-tourism, which can enhance the social, environmental, and economic status of a territory, plays an important role in the sustainable tourism initiatives (Lumsdon 2000). The development of cycle-tourism can help create job opportunities like the cycle-tourism guide to satisfy the demand of cycle-tourists, promote a sustainable mode of transportation, improve the quality of life and wellness of local residents, and preserve the local environment by reducing pollution, etc. (Rode et al. 2017; Ritchie 1998; Wearing and Neil 2009; Lamont 2009). The cycle-tourism has already been well developed in Europe in both rural and non-rural regions, like Stockholm and Copenhagen. As a form of slow tourism, cycle-tourism allows tourists to experience a territory with the conservation of the environmental and social dimensions of life, enhancing tourists' quality of time on the trip and engagement with local culture, people, and places (Gazzola et al. 2018; Lumsdon and McGrath 2011).

Many researches have explored the field of cycle-tourism. Chen and Chen (2013) examined the preference of both cycle-tourists and recreational cyclists for cycling routes in Taiwan with the stated preference method. The study suggested that cycling routes with attractions, basic facilities, tourist information centers, and bike paths along the way are preferred. Specifically, cycle-tourists are mainly motivated by the willingness to enjoy attractions and engage in sightseeing. Deenihan and Caulfield (2015) examined how different types of cycling infrastructure are valued by cycle-tourists with intercept stated preference survey. Han, Meng, and Kim (2017) investigated travelers' decision process to decide on cycle-tourism as a form of sustainable tourism activity, in consideration of the moderating impact of other unsustainable alternatives' attractiveness. Gazzola et al. (2018) explored the characteristics of cycle-tourism development in northern Italy to determine the links which exist between sustainable tourism and the cycle-tourists who would like to discover little-known or remote territories. As cycle-tourists often require additional car journey to the location where their cycling will take place, Chen and Cheng (2016) studied the integrated bike-rail transport service and cyclists' preferences of the service.

Cycle-tourist route planning and optimization

Although the concept of cycle-tourism has been widely discussed in many recent literatures, there have been very few researches on the optimization of the cycle-tourist route-planning problem. Souffriau et al. (2011); Verbeeck, Vansteenwegen, and Aghezzaf (2014) formulated the model as arc orienteering problems to optimize the total score or profit on the recreational cycling route with a constraint on the total length of the route. Černá et al. (2014); Malucelli, Giovannini, and Nonato (2015) designed the routes for cycle-tourists by maximizing the attractiveness of the route subject to the monetary budget and time constraint. However, to the authors' best knowledge, no study has formulated a multi-objective optimization model to cater to the various needs of cycle-tourists.

Discussion

In this paper, we formulated a multi-objective cycle-tourist route-planning model that aims to better satisfy the requirements of cycle-tourists and promote cycle-tourism. The four objective functions or key indicators are selected to cater to various cycle-tourists' preferences. Related literatures have also been reviewed to choose the key indicators for cycle-tourist route planning. First, the objective to maximize cycle-tourists' utility of visiting POIs is considered as the maximization of tourist satisfaction is the main objective in the tourist route-planning problem (Gavalas et al. 2014). Second, the objective to minimize the travel cost or travel time of cycle-tourist is also considered when planning the trip, as tourists seek to minimize the travel time in order to optimize the time spent at attraction points (Gavalas et al. 2015). Third, we take the objective to maximize the Bicycle Level of Service (BLOS) into account which measures the cycling comfort level provided to the bicyclists or cycle-tourists on the cycling route taken. The BLOS concept affects the comfort perception and route choice of bicyclists, which has been addressed in many recent literatures (Asadi-Shekari et al., 2013; Bl, Andrášik, and Kubeček 2015; Calvey et al. 2015; Bai et al. 2017). Lastly, bicyclists or cycle-tourists prefer connected bikeway routes to fragmented ones as intersections pose safety risks and additional travel time, hence the objective to minimize the number of intersections on the cycle-tour is also introduced (Lin and Yu 2013; Lin and Liao 2016).

Augmented ∊-constraint method (AUGMECON) is applied to solve the multi-objective mixed-integer linear optimization problem for cycle-tourist route planning, where a set of nondominated cycle-tour route design alternatives are generated and the trade-offs between different objectives are addressed. Two numerical examples are used to demonstrate the feasibility of the model formulated and the solution algorithm for cycle-tourist route planning.

Methodology

The research problem is to optimize the cycle-tourist route-planning problem by selecting the POIs to visit and determine the visiting sequence, in order to maximize cycle-tourists' utility of visiting POIs, minimize the total travel time, maximize the BLOS, and minimize the number of intersections on the cycling route taken due to safety concerns, subject to constraints like monetary and time budget.

Cycle-tourist route-planning model

Assumption

It is assumed that the attractiveness of each POI i, i.e. the score/profit p_i for personal trip design, is determined before model analysis, which depends on different fields of interest (Gunawan, Lau, and Vansteenwegen 2016). The way of computing p_i should be based on information retrieval techniques, for example, as in Kotiloglu et al. (2017) and Souffriau et al. (2008), etc.

Model formulation

Let G be a complete undirected graph $G = (V, E)$, where V is the node set, E is the set of edges or transportation links. $V^a = \{1, 2, \ldots, n\}$ is the set of POIs to be visited, and node 1 and node n denote the starting and ending point of each cycle-tour, respectively, $V^a \in V$. V^c represents the node set excluding the starting and ending points with $V^c = V^a\{1, n\}$. Note that the starting node 1 and ending node n can refer to the same POI location such as a certain hotel or different locations such as a restaurant, MRT station or different hotels. The cycle-tour can be seen as a Hamiltonian circuit in the first case and as a Hamiltonian path in the second case where each POI is visited at most once.

Based on the model framework and definitions, the cycle-tourist route-planning problem is formulated as below:

$$max \quad \sum_{i=1}^{n-1}\sum_{j=2}^{n} p_i \cdot x_{ij} \tag{1}$$

$$min \quad \sum_{i=1}^{n-1}\sum_{j=2}^{n} c_{ij} \cdot x_{ij} \tag{2}$$

$$min \quad \sum_{i=1}^{n-1}\sum_{j=2}^{n} s_{ij} \cdot x_{ij} \cdot d_{ij} \tag{3}$$

$$min \quad \sum_{i=1}^{n-1}\sum_{j=2}^{n} I_{ij} \cdot x_{ij} \tag{4}$$

$$\sum_{i=2}^{n} x_{1i} = \sum_{j=1}^{n-1} x_{jn} = 1 \tag{5}$$

$$\sum_{i=2}^{n} x_{i1} = \sum_{j=1}^{n-1} x_{nj} = 0 \tag{6}$$

$$\sum_{i=1}^{n-1} x_{ik} = \sum_{j=2}^{n} x_{kj} \leq \in_k, \quad k \in V^c, k \neq i, j \tag{7}$$

$$\sum_{i=1}^{n-1}\sum_{j=2}^{n} c_{ij} \cdot x_{ij} + \sum_{i=2}^{n-1}\sum_{j=2}^{n} c_i^v \cdot x_{ij} \leq T \tag{8}$$

$$\sum_{i=1}^{n-1}\sum_{j=2}^{n} e_i \cdot x_{ij} \leq B \tag{9}$$

$$2 \leq u_i \leq n, \quad i \in \{1, 2, \ldots, n\} \tag{10}$$

$$u_i - u_j + 1 \leq (n-1)(1 - x_{ij}), \quad i, j \in \{2, \ldots, n\} \tag{11}$$

$$x_{ij} \in \{0, 1\} \tag{12}$$

Equation 1 is the first objective, which aims to maximize the total collected profit. The binary variable x_{ij} indicate whether POI i is visited after attraction j, $i \neq j$, on the path from node 1 to node n. The profit value p_i can be seen as the utility of visiting the particular POI i compared to others (Kotiloglu et al. 2017; Deshpande and Karypis 2004). According to the interests of various tourists, information retrieval, or data mining techniques are applied to calculate p_i for each POI i (Souffriau et al. 2008; Gavalas et al. 2015). Equation 2 is the second objective which aims to minimize the total travel time, which should take both the link travel time and the intersection delay into account. Equation 3 is the third objective function, which aims to maximize the Bicycle Level of Service (BLOS) condition for traveling among POIs. BLOS is defined as a measure for evaluating the quality of service provided by cycling facilities in the streets, affecting the comfort perception, mode choice, and route choice of cyclists (Asadi-Shekari et al., 2013). In this paper, similar to Klobucar and Fricker (2007) and Lowry et al. (2012), it is assumed that the bicyclists would like to choose the route in order to minimize the traveling distance times the BLOS score. d_{ij} represents the traveling distance from the POI location i to j. s_{ij} represents the BLOS score on road link (i, j). The BLOS score of road link is jointly affected by multiple performance measures, such as traffic conditions, facility characteristics, environment factors, etc. (Zhu and Zhu 2019a; Bai et al. 2017; Kang and Lee 2012; Yamanaka and Namerikawa 2007; Jones and Carlson 2003). Typically, each attribute of the bike-way link is awarded a certain number of points, which are then summed up to derive a BLOS score which categorizes the comfort and safety level of the cycling facility from the most desirable to the most undesirable. There exist different methods for BLOS evaluation with various choices of cycling facility attributes, various point systems, and various ways of combining the points, supported by theory and empirical findings based on the local context (Lowry et al. 2012). Equation 4 is the fourth objective which aims to minimize the number of intersections on the cycle-tour among POIs selected due to safety concerns and the desire to minimize the number of interruptions incurred to cycle-tourists, where I_{ij} represents the number of intersections from POI i to j.

For the constraints, Equation 5 and 6 ensure that the cycle-tour starts from node 1 and end at node n. Equation 7 ensures that the path starting at node 1 and ending at node n is connected and all POIs are visited at most once. Moreover, only the available POI for each period is taken into consideration, where the binary variable ε_i is introduced, taking the value of 1 if POI i is available on the day of visiting and 0 otherwise. Equation 8 handles the time budget T for each cycle-tour, where c_i^v represents the estimated visiting time at POI i. Equation 9 bounds the total monetary budget B of each cycle-tour. A visiting cost e_i which includes the entrance fee, food price, etc., is defined for each POI. Equation 10 and Equation 11 are subtour elimination constraints where u_i represents the place of node i in the path from 1 to n. Equation 12 is a binary constraint for decision variable x_{ij}.

Algorithms

Multi-objective optimization

According to Deb (2014), the principle for an ideal multi-objective optimization procedure consists of two steps: the first step is to find multiple trade-off optimal solutions with different values for the objectives; the second step is to select one of the solutions computed in the first step with higher-level information. Decision makers who are able to express preferences for solutions, or a cycle-tourist in the case of cycle-tourist route-planning problem will be involved in selecting a particular desirable solution out of the Pareto-optimal set, i.e. the nondominated set of the entire feasible decision space.

To evaluate the goodness of solutions of the multi-objective optimization problem, the concept of dominance is introduced. For the multi-objective optimization problem with S objective functions, a solution x' is defined to dominate another solution x ($x'x$) if the following two conditions are met:

1. x' is no worse than x for all objectives, i.e. $f_s(x') \leq f_s(x)$ for all objective functions to be minimized and $f_s(x') \geq f_s(x)$ for all objective functions to be maximized, $s \in \{1, 2, \ldots, S\}$;

2. x' is strictly better than x for at least one objective, i.e. $f_{\bar{s}}(x') < f_{\bar{s}}(x)$ for all objective functions to be minimized and $f_{\bar{s}}(x') > f_{\bar{s}}(x)$ for all objective functions to be maximized, $\exists \bar{s} \in \{1, 2, \ldots, S\}$.

A solution x^* is defined as a non-dominated solution or Pareto-optimal solution if and only if there is no other solution x that dominates x^*. Furthermore, we also have the definition of the weakly non-dominated solution, when the conditions on the non-dominated solution are relaxed. A feasible solution x^* is defined as a weakly nondominated solution if there is no other feasible solution x which only satisfies Equation 13 for the objective functions to be minimized or Equation 14 for the objective functions to be maximized. Weakly nondominated solutions are usually not desired in the multi-objective optimization problem since they may actually be dominated by other nondominated solutions, hence more preferred Pareto-optimal solutions or nondominated solutions are usually pursued by rational decision makers.

$$f_s(x) < f_s(x^*), \quad s \in \{1, 2, \ldots, S\} \tag{13}$$

$$f_s(x) > f_s(x^*), \quad s \in \{1, 2, \ldots, S\} \tag{14}$$

Augmented ∈-constraint method–AUGMECON

To overcome the sensitivity issues and optimality issues of the traditional weighted-sum method commonly used in the literature when solving a multi-objective optimization problem, augmented ∈-constraint method is adopted in this paper. The method preserves one of the objective functions which is regarded as the most significant one while the rest of the objective functions are converted to constraints, which are restricted within user-specific ∈ values (Mavrotas 2009). No weight combination needs to be selected as a set of nondominated cyclist-tourist route-planning solutions can be explicitly generated by the augmented ∈-constraint method. The augmented ∈-constraint method (AUGMECON) adopted in this paper avoids the generation of weakly nondominated solutions which can be obtained with the traditional ∈-constraint method and speeds up the algorithm by avoiding redundant iterations (Mavrotas 2009). With AUGMECON, the original multi-objective optimization formulation $[P1]$ (as shown in Equation 15) is converted to $[P2]$ (as shown in Equation 16), in which the first objective function $f_1(x)$ for the maximization of total collected profit during the trip is retained to be solved. Slack variables sl_2, sl_3, sl_4 are introduced to demonstrate the inequality constraints $f_s(x) \leq \in_s, \quad s = \{2, 3, 4\}$.

$$[P1] \quad \underset{x \in X}{max\, f_1(x); min\, f_2(x); min\, f_3(x); min\, f_4(x);} \tag{15}$$

$$\begin{aligned} &[P2] \;\; max\, f_1(x) + eps * (sl_2/r_2 + sl_3/r_3 + sl_4/r_4); \\ &s.t.\, f_2(x) + sl_2 = \in_2, \\ &f_3(x) + sl_3 = \in_3, \\ &f_4(x) + sl_4 = \in_4, \\ &x \in X \end{aligned} \tag{16}$$

where eps is an adequately small value normally from 10^{-6} to 10^{-4}, and r_2, r_3, r_4 represent the ranges of objectives 2, 3, 4, respectively, which are introduced for normalization. $\in_s$ ($s \in \{2, 3, 4\}$) represents an upper bound of objective function $f_s(x)$ ($s \in \{2, 3, 4\}$), where $\in_s = f_s^-(x) + b/(B-1) \cdot (f_s^+(x) - f_s^-(x)), \quad s \in \{2, 3, 4\}$. $f_s^-(x)$ and $f_s^+(x)$ represents the minimum possible value and maximum possible value which can be taken by objective function $f_s(x)$ respectively. With the parametrical variations of $\in_s$ values, Equation 16 is solved repeatedly for different $b \in \{1, 2, \ldots, B\}$, which can provide at most 3^B nondominated solutions. Algorithm 1 elaborates the process of computing the Pareto-optimal set $\hat{P}$. $\hat{f}_2(x), \hat{f}_3(x), \hat{f}_4(x)$ represent sets of different values that can be taken by objective functions $f_2(x), f_3(x), f_4(x)$, respectively, and the elements in the three sets are denoted by $f_2^p(x)$, $f_3^q(x)$, $f_4^g(x)$, respectively. In each iteration, $\in$ value pair $(\in_2, \in_3, \in_4)$ for which $[P2]$ has a feasible solution is solved for optimality, generating an optimal solution $x^* = opt((f_1(x) + eps * (sl_2/r_2 + sl_3/r_3 + sl_4/r_4)), \in_2, \in_3, \in_4)$, and the solution x^* is a Pareto-optimal solution or non-dominated solution if it has not been generated in the previous iterations (Abounacer, Rekik, and Renaud 2014).

Algorithm 1 Augmented $\in$-constraint method

$$\hat{P} \leftarrow \emptyset$$

for p=1 to $|\hat{f}_2(x)|$ **do**
for q=1 to $|\hat{f}_3(x)|$ **do**
for g=1 to $|\hat{f}_4(x)|$ **do**

$$\in_2 = \hat{f}_2^p(x),\ \in_3 = \hat{f}_3^q(x),\ \in_4 = \hat{f}_4^g(x)$$

while $P(\in_2, \in_3, \in_4)$ has a feasible solution **do**

$$\in_2^* \leftarrow \in_2,\ \in_3^* \leftarrow \in_3,\ \in_4^* \leftarrow \in_4$$

;

$$x^* \leftarrow opt((Z_1 + eps * (sl_2/r_2 + sl_3/r_3 + sl_4/r_4)), \in_2^*, \in_3^*, \in_4^*)$$

$$\hat{P} \leftarrow \hat{P} \cup \{x^*\}$$

end
end
end
end
for $x^* \in \hat{P}$ **do**
if $x^* = \hat{x}$, for $\hat{x} \in \hat{P}$, **then**

$$\hat{P} \leftarrow \hat{P} \backslash \{x^*\}$$

end
end
return $\hat{P}$

We have the following proof that Equation 16 only produces nondominated solutions or Pareto optimal solutions, according to Mavrotas (2009).

Proof. Suppose by contradiction that an alternative optimal solution x' exists for Equation 15 which dominates the optimal solution x^* which is obtained from Equation 16, so that

$$\begin{aligned} f_1(x') &\geq f_1(x^*), \\ f_2(x') &\leq f_2(x^*), \\ f_3(x') &\leq f_3(x^*), \\ f_4(x') &\leq f_4(x^*), \end{aligned} \tag{17}$$

with at least one strict inequality, hence

$$\begin{aligned} \in_2 - sl'_2 &\leq \in_2 - sl_2^*, \\ \in_3 - sl'_3 &\leq \in_3 - sl_3^*, \\ \in_4 - sl'_4 &\leq \in_4 - sl_4^*, \end{aligned} \tag{18}$$

with at least one strict inequality. Then we have

$$f_1(x') + sl'_2/r_2 + sl'_3/r_3 + sl'_4/r_4 \geq f_1(x^*) + sl_2^*/r_2 + sl_3^*/r_3 + sl_4^*/r_4 \tag{19}$$

With Equation 19, we find that the initial assumption which suggests that x^* is an optimal solution of Equation 16 is contradicted. Therefore, we conclude that x^* generated from Equation 16 is a non-dominated solution.□

To sum up, Figure 1 shows an overview of the algorithm in the cycle-tourist route design problem. Initially, the transportation network and related parameters are loaded for the study area. Then the multi-objective optimization model is formulated to cater for four different objective functions for cycle-tourist route planning. The solution algorithm search for the Pareto-optimal solutions to the multi-objective optimization model, and nondominated cycle-tourist route plans are derived if the plan has not been generated in previous iterations.

Numerical studies

Numerical example 1

For ease of illustration, we use a six-node case as an example, where the starting node 1 is numbered as 1, the ending node n is numbered as 6. Random numbers are given to different characteristics of POIs and trips between each two of the POIs. Two nondominated cycle-tourist trip planning alternatives derived with augmented $\in$-constraint method are summarized in Table 2. The objective values and performances of Z_1, Z_2, Z_3, Z_4 are summarized in Table 3 to address the trade-offs between the maximization and minimization of the four objective functions Z_1, Z_2, Z_3, Z_4. Equation 20 and Equation 21 are applied to measure the performance of the minimization and maximization of objectives, respectively (Lin and Yu 2013; Zhu and Zhu 2019b):

$$performance = (Max - OV)/(Max - Min) \tag{20}$$

$$performance = (OV - Min)/(Max - Min) \tag{21}$$

where Max is the maximum value of a certain objective function, Min is the minimum value of a certain objective function, and OV represents a particular non-dominated solution's objective value.

In the first column of Table 3, since the first objective function Z_1 to maximize the total collected profit, Equation 21 should be applied for performance evaluation. In this case, Max=5 and

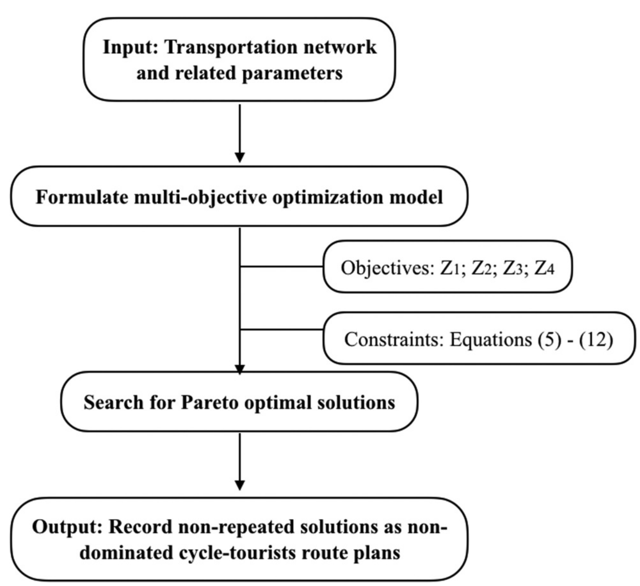

Figure 1. Overview flowchart.

Table 1. Notations.

Symbol	Definition
G	Complete undirected graph G = (V,E)
V	Set of nodes or POIs to be visited
E	Set of edges, such that $A = \{(i,j) \in V \times V : i \neq j\}$
V^a	The set of POIs to be visited, $V^a = \{1, 2, \ldots, n\}$, $V^a \in V$
1	The starting node of each cycle-tour
n	The ending node of each cycle-tour
V^c	The node set excluding the starting and ending nodes with $V^c = V^a\{1, n\}$
p_i	Profit of POI i, which can be seen as the utility of visiting it compared to other POIs
$\in_i$	Binary variable, which takes the value of 1 if POI i is available on the day of visiting and 0 otherwise
c_{ij}	Cost of traveling from i to j, $(i,j) \in E$, including link travel time and intersection delay
s_{ij}	Bicycle suitability score on transportation link (i,j), $(i,j) \in E$
d_{ij}	Traveling distance from i to j, $(i,j) \in E$
I_{ij}	Number of intersections from i to j, $(i,j) \in E$
T	The travel time budget of cycle-tourist
c_i^v	The visiting time at POI i
e_i	Visiting cost for POI i (e.g. entrance fees and food prices)
B	Total monetary budget for cycle-tourist
x_{ij}	Binary variable, which takes the value of 1 if POI j is visited after attraction i, $i \neq j$; and 0 otherwise, for every path from 1 to n
u_i	The place of node i in the path from 1 to n

Table 2. Result of the 6-node network.

link(i,j), (j,i)	x_{ij}^1, x_{ji}^1	x_{ij}^2, x_{ji}^2
(1,2), (2,1)	0, 0	0, 0
(1,3), (3,1)	0, 0	0, 0
(1,4), (4,1)	0, 0	0, 0
(1,5), (5,1)	0, 0	1, 0
(1,6), (6,1)	1, 0	0, 0
(2,3), (3,2)	0, 0	0, 0
(2,4), (4,2)	0, 0	0, 0
(2,5), (5,2)	0, 0	0, 0
(2,6), (6,2)	0, 0	0, 0
(3,4), (4,3)	0, 0	0, 0
(3,5), (5,3)	0, 0	0, 0
(3,6), (6,3)	0, 0	0, 0
(4,5), (5,4)	0, 0	0, 0
(4,6), (6,4)	0, 0	0, 0
(5,6), (6,5)	0, 0	1, 0

${}^a\delta^l_{ijk}$ represents the l^{th} nondominated solution

Min=0. The objective value (OV) for Z_1 is 0 for plan 1 and 5 for plan 2. Based on Equation 21, the performance value of plan 1 is calculated as (0–0)/(5–0) = 0 and the performance value of plan 2 is calculated as (5–0)/(5–0) = 100%. In the third column of Table 3, Equation 20 should be applied for performance evaluation, as the third objective function Z_3 is for minimization. In this case, Max=7; Min=5; the objective value (OV) for Z_3 is 5 for plan 1 and 7 for plan 2. According to Equation 20, the performance value of plan 1 is calculated as (7–5)/(7–5) = 100% and the performance value of plan 2 is calculated as (7–7)/(7–5) = 0. For objective Z_2 and Z_3, since there is no difference in the performance of plan 1 and plan 2, 100% is simply indicated for both plans.

Table 3. Objective values of nondominated solutions.

Nondominated	Objective values and performances			
Alternative plans	Z_1	Z_2	Z_3	Z_4
Plan 1	0	9	5	9
	(0%)	(100%)	(100%)	(100%)
Plan 2	5	9	7	9
	(100%)	(100%)	(0%)	(100%)

The result shows that cycle-tourist route plan 2 outperforms plan 1 in terms of the first objective function to maximize cycle-tourists' utility of visiting POIs. On the other hand, cycle-tourist route plan 1 outperforms plan 2 regards to the third objective to maximize the BLOS on the cycling route taken. The nondominated plans have similar performances in terms of the second and the fourth objective functions. Individual cycle-tourist can select the desired cycle-tour route plan out of the table based on his or her own preference and judgment according to the performance of different objectives in each non-dominated route plan. In this simple numerical example, plan 1 will be highly recommended to the cycle-tourists who indicate that they regard cycling comfort level along the cycling route as important. On the other hand, plan 2 will be highlighted to the cycle-tourists who value the attractiveness of the points of interest to visit.

Numerical example 2

For the second numerical example, we consider 15 most popular POIs and Landmarks which are within cycling distance near the downtown area of Singapore according to *TripAdvisor*, and two hotels are considered as the origin and destination of the cycle-tour in this numerical example, as shown in Figure 2 (Trip Advisor 2019). To store the information related to each POI i, we create Table 4 following a method similar to the one used by Abbaspour and Samadzadegan (2011). The attractiveness of each POI i, i.e. the score/profit for personal trip design p_i, is determined with the ratings on the *TripAdvisor* first on a scale from 1 to 5. Individual cycle-tourist can modify the p_i data in order to reflect their priority in visiting each POI and the suggested staying time according to personal interest. If the cycle-tourist leaves the table unchanged, we apply the default setting of the table. The value of I_{ij} and d_{ij} are calculated directly from Google Map, and the value of c_{ij} can be estimated with cycling speed 15 km/h. The value of the BLOS score s_{ij} should be calculated with the existing evaluation method, such as HCM (2010). However, since not all data required for the BLOS evaluation method is available in the

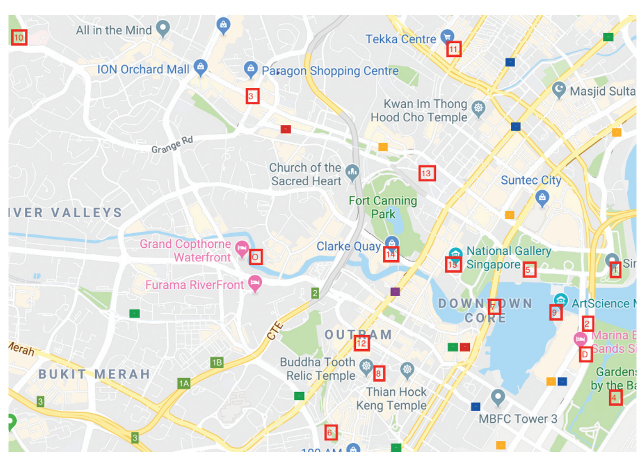

Figure 2. Popular POIs selected according to *TripAdvisor*.

Table 4. Information on POIs.

POI ID	POI name	Priority (p_i)	Suggested staying time (c_i^v)	Visiting cost (e_i)
1	Singapore Flyer	4.5	2.5 h	33 USD
2	Marina Bay Sands Skypark	4.5	1.5 h	21 USD
3	Supertree Grove	4.5	1.5 h	23 USD
4	Orchard Road	4	4 h	0

context of Singapore, for simplification purpose, we estimate the BLOS based on the build environment with hypothetical values in this study. Moreover, we assume that $\in_k = 1$ for all POIs. The monetary budget B is set to 25 Singapore dollar and the time budget is set to 6.5 hours.

With the augmented $\in$-constraint method (AUGMECON), five non-dominated solutions are generated with a running time of 10.76 seconds when the value of B is set to 3. The alternative cycle-tour route plans are summarized in Figure 3.

The trade-offs between total profit collected from visiting POIs, the total travel time, the Bicycle Level of Service (BLOS) during the trip, the number of intersections are addressed and summarized in Table 5. Plan 1 performs the best in terms of the second objective function to minimize the total travel time, but has the worst performance in the first objective function to maximize cycle-tourists' utility of visiting POIs during the trip. Plan 2 and plan 3 have the best performance in the maximization of BLOS on the cycling route taken by the cycle-tourists, demonstrating a better cycling comfort level. Plan 3 and plan 4 provide the minimal number of intersections on the cycling route taken, which provides a more connected cycling route rather than a fragmented one. Plan 5 performs the best when maximizing cycle-tourists' utility of visiting POIs, but has the worst performances in the other objective functions in comparison with other nondominated cycle-tourist route plans generated.

Refer to Table 5 and the discussion above on the trade-offs among various objective functions, cycle-tourist can select the

Table 5. Multiple objectives trade-offs.

Nondominated	Objective values and performances			
Alternative plans	Z_1	Z_2	Z_3	Z_4
Plan 1	0 (0%)	0.26 (100%)	5.85 (84.2%)	22 (54.2%)
Plan 2	4.5 (17.3%)	0.2867 (87.5%)	4.3 (100%)	14 (87.5%)
Plan 3	4 (15.4%)	0.2867 (87.5%)	4.3 (100%)	11 (100%)
Plan 4	12.5 (48.1%)	0.3133 (75.0%)	5.35 (89.3%)	11 (100%)
Plan 5	26 (100%)	0.4733 (0%)	14.1 (0%)	35 (0%)

(a) Plan 1 (b) Plan 2

(c) Plan 3 (d) Plan 4

(e) Plan 5

Figure 3. Alternative cycle-tour plans.

cycle-tour routing alternative according to their preference. Additional filtering technique may also be applied to reduce the number of alternatives presented to cycle-tourists, such that cycle-tourists can choose their route more conveniently. For example, Plan 1 which connects origin hotel to destination hotel directly without visiting any POIs is not reasonable, hence can be eliminated. Plan 2 and plan 3 will be highlighted to the cycle-tourists if they indicate that they regard cycling comfort level along the route as more important. Plans 3 and 4 will be highly recommended to the cycle-tourists who value safety level and continuity of cycling route. Plan 5 will be highlighted to the cycle-tourists who think highly of the points of interest to visit.

Conclusions

As a type of niche and sustainable tourism, cycle-tourism is increasingly promoted in recent years, but there are still few researches on the cycle-tourist route-planning problem. Different from tourists traveling with auto-mobiles who are most concerned about the utility of visiting POIs and the travel time, the cycling comfort and safety level on is also significant to cycle-tourists. To address the research gap, in this paper, a multi-objective mixed-integer linear programming model is formulated to optimize the cycle-tourist route-planning problem, which better satisfies the needs of cycle-tourists in order to promote cycle-tourism. We consider the characteristics of cycle-tourists and propose a multi-objective cycle-tourist route-planning problem, taking cycle-tourist's utility of visiting attraction points, the travel time, the Bicycle Level of Service (BLOS), and the number of intersections on the cycling route into account, subject to constraints like monetary and time budget. Augmented $\in$-constraint method (AUGMECON) has been applied to generate the nondominated cycle-tourist route plans, such that cycle-tourists can select the route based on their personal route preferences.

This paper can help plan cycle-tourists' trips and serve as the basis for developing real-world mobile applications for cycle-tourist route planning, where users need to specify the origin and destination locations, as well as the route preferences. The method can also be applied to a larger scale transportation network that contains more attraction points. The paper may further provide insight for tourism-related bike-way network design, according to the routing plans selected by cycle-tourists.

Disclosure statement

No potential conflict of interest was reported by the author.

References

Abbaspour, R. A., and F. Samadzadegan. 2011. "Time-dependent Personal Tour Planning and Scheduling in Metropolises." *Expert Systems with Applications* 38: 12439–12452.

Abounacer, R., M. Rekik, and J. Renaud. 2014. "An Exact Solution Approach for Multi-objective Location–transportation Problem for Disaster Response." *Computers & Operations Research* 41: 83–93.

Asadi-Shekari, Z., M. Moeinaddini, and M. Zaly Shah. 2013. "Non-motorised Level of Service: Addressing Challenges in Pedestrian and Bicycle Level of Service." *Transport Reviews* 33: 166–194.

Bai, L., P. Liu, C. Y. Chan, and Z. Li. 2017. "Estimating Level of Service of Mid-block Bicycle Lanes considering Mixed Traffic Flow." *Transportation Research Part A: Policy and Practice* 101: 203–217.

Bl, M., R. Andrášik, and J. Kubeček. 2015. "How Comfortable are Your Cycling Tracks? A New Method for Objective Bicycle Vibration Measurement." *Transportation Research Part C: Emerging Technologies* 56: 415–425.

Broach, J., J. Dill, and J. Gliebe. 2012. "Where Do Cyclists Ride? A Route Choice Model Developed with Revealed Preference Gps Data." *Transportation Research Part A: Policy and Practice* 46: 1730–1740.

Calvey, J., J. Shackleton, M. Taylor, and R. Llewellyn. 2015. "Engineering Condition Assessment of Cycling Infrastructure: Cyclists' Perceptions of Satisfaction and Comfort." *Transportation Research Part A: Policy and Practice* 78: 134–143.

Černá, A., J. Černỳ, F. Malucelli, M. Nonato, L. Polena, and A. Giovannini. 2014. "Designing Optimal Routes for Cycle-tourists." *Transportation Research Procedia* 3: 856–865.

Chen, C. F., and P. C. Chen. 2013. "Estimating Recreational Cyclists' Preferences for Bicycle routes–Evidence from Taiwan." *Transport Policy* 26: 23–30.

Chen, C. F., and W. C. Cheng. 2016. "Sustainability Si: Exploring Heterogeneity In Cycle Tourists ' preferences for an integrated bike-rail transport service." *Networks and Spatial Economics* 16: 83–97.

Deb, K. 2014. *Multi-objective Optimization, In: Search Methodologies*, 403–449. Springer.

Deenihan, G., and B. Caulfield. 2015. "Do Tourists Value Different Levels of Cycling Infrastructure?" *Tourism Management* 46: 92–101.

Deshpande, M., and G. Karypis. 2004. "Item-based Top-n Recommendation Algorithms." *ACM Transactions on Information Systems (TOIS)* 22: 143–177.

Gavalas, D., V. Kasapakis, C. Konstantopoulos, G. Pantziou, N. Vathis, and C. Zaroliagis. 2015. "The eCOMPASS Multimodal Tourist Tour Planner." *Expert Systems with Applications* 42: 7303–7316.

Gavalas, D., C. Konstantopoulos, K. Mastakas, and G. Pantziou. 2014. "A Survey on Algorithmic Approaches for Solving Tourist Trip Design Problems." *Journal of Heuristics* 20: 291–328.

Gazzola, P., E. Pavione, D. Grechi, and P. Ossola. 2018. "Cycle Tourism as a Driver for the Sustainable Development of Little-known or Remote Territories: The Experience of the Apennine Regions of Northern Italy." *Sustainability* 10: 1863.

Gunawan, A., H. C. Lau, and P. Vansteenwegen. 2016. "Orienteering Problem: A Survey of Recent Variants, Solution Approaches and Applications." *European Journal of Operational Research* 255: 315–332.

Han, H., B. Meng, and W. Kim. 2017. "Emerging Bicycle Tourism and the Theory of Planned Behavior." *Journal of Sustainable Tourism* 25: 292–309.

HCM. 2010. "Highway Capacity Manual. Transportation Research Board." National Research Council: Washington, D.C

Hochmair, H. 2005. "Towards a Classification of Route Selection Criteria for Route Planning Tools." In *Developments in Spatial Data Handling*, 481–492. Springer.

Hochmair, H. H., and Z. J. Fu. 2009. *Web Based Bicycle Trip Planning for Broward County*. Florida.

Hood, J., E. Sall, and B. Charlton. 2011. "A GPS-based Bicycle Route Choice Model for San Francisco, California." *Transportation Letters* 3: 63–75.

Hrnčř, J., P. Žileckỳ, Q. Song, and M. Jakob. 2017. "Practical Multicriteria Urban Bicycle Routing." *IEEE Transactions on Intelligent Transportation Systems* 18: 493–504.

Hyde, K. F., and R. Lawson. 2003. "The Nature of Independent Travel." *Journal of Travel Research* 42: 13–23.

Jones, E., and T. Carlson. 2003. "Development of Bicycle Compatibility Index for Rural Roads in Nebraska. Transportation Research Record." *Journal of the Transportation Research Board* 124–132.

Kang, K., and K. Lee. 2012. "Development of a Bicycle Level of Service Model from the User' S Perspective." *KSCE Journal of Civil Engineering* 16: 1032–1039.

Klobucar, M. S., and J. D. Fricker. 2007. "Network Evaluation Tool to Improve Real and Perceived Bicycle Safety." *Transportation Research Record* 2031: 25–33.

Kotiloglu, S., T. Lappas, K. Pelechrinis, and P. Repoussis. 2017. "Personalized Multi-period Tour Recommendations." *Tourism Management* 62: 76–88.

Lamont, M. 2009. "Reinventing the Wheel: A Definitional Discussion of Bicycle Tourism." *Journal of Sport & Tourism* 14: 5–23.

Lee, C. F. 2014. "An Investigation of Factors Determining Cycling Experience and Frequency." *Tourism Geographies* 16: 844–862.

Lin, J. J., and R. Y. Liao. 2016. "Sustainability SI: Bikeway Network Design Model for Recreational Bicycling in Scenic Areas." *Networks and Spatial Economics* 16: 9–31.

Lin, J. J., and C. J. Yu. 2013. "A Bikeway Network Design Model for Urban Areas." *Transportation* 40: 45–68.

Lowry, M., D. Callister, M. Gresham, and B. Moore. 2012. "Using Bicycle Level of Service to Assess Community-wide Bikeability." In *91st Annual Meeting of the Transportation Research Board*. Washington, DC: Transportation Research Board.

Lumsdon, L. 2000. "Transport and Tourism: Cycle Tourism–a Model for Sustainable Development?" *Journal of Sustainable Tourism* 8: 361–377.

Lumsdon, L. M., and P. McGrath. 2011. "Developing A Conceptual Framework for Slow Travel: A Grounded Theory Approach." *Journal of Sustainable Tourism* 19: 265–279.

Majumdar, B. B., S. Mitra, and P. Pareekh. 2019. "On Identification and Prioritization of Motivators and Deterrents of Bicycling." *Transportation Letters,* 1–13.

Malucelli, F., A. Giovannini, and M. Nonato. 2015. "Designing Single Origin-destination Itineraries for Several Classes of Cycle-tourists." *Transportation Research Procedia* 10: 413–422.
Mavrotas, G. 2009. "Effective Implementation of the ε-constraint Method in Multi-objective Mathematical Programming Problems." *Applied Mathematics and Computation* 213: 455–465.
Ritchie, B. W. 1998. "Bicycle Tourism in the South Island of New Zealand: Planning and Management Issues." *Tourism Management* 19: 567–582.
Rode, P., G. Floater, N. Thomopoulos, J. Docherty, P. Schwinger, A. Mahendra, and W. Fang. 2017. "Accessibility in Cities: Transport and Urban Form." In *Disrupting Mobility*, 239–273. Springer.
Sener, I. N., N. Eluru, and C. R. Bhat. 2009. "An Analysis of Bicycle Route Choice Preferences in Texas, US." *Transportation* 36: 511–539.
Song, Q., P. Zilecky, M. Jakob, and J. Hrncir, 2014. "Exploring Pareto Routes in Multi-criteria Urban Bicycle Routing." In Intelligent Transportation Systems (ITSC), 2014 IEEE 17th International Conference on, IEEE. 1781–1787.
Souffriau, W., P. Vansteenwegen, G. V. Berghe, and D. Van Oudheusden. 2011. "The Planning of Cycle Trips in the Province of East Flanders." *Omega* 39: 209–213.
Souffriau, W., P. Vansteenwegen, J. Vertommen, G. V. Berghe, and D. V. Oudheusden. 2008. "A Personalized Tourist Trip Design Algorithm for Mobile Tourist Guides." *Applied Artificial Intelligence* 22: 964–985.
Su, J. G., M. Winters, M. Nunes, and M. Brauer. 2010. "Designing a Route Planner to Facilitate and Promote Cycling in Metro Vancouver, Canada." *Transportation Research Part A: Policy and Practice* 44: 495–505.
Szeto, W. 2016. "Guest Editorial: Special Issue on Quantitative Approaches to Environmental Sustainability in Transportation Networks." *Networks and Spatial Economics* 16: 1–8.
Szeto, W., X. Jaber, and S. Wong. 2012. "Road Network Equilibrium Approaches to Environmental Sustainability." *Transport Reviews* 32: 491–518.
Trip Advisor, 2019. "Things to Do in Singapore." html#ATTRACTION_SORT_WRAPPER. Online; accessed 01 February 2019. https://www.tripadvisor.com.sg/Attractions-g294262-Activities-Singapore
Turverey, R. J., D. D. Cheng, O. N. Blair, J. T. Roth, G. M. Lamp, and R. Cogill. 2010. "Charlottesville Bike Route Planner." In Systems and Information Engineering Design Symposium (SIEDS), 2010 IEEE, IEEE. pp. 68–72.
Vansteenwegen, P., and D. Van Oudheusden. 2007. "The Mobile Tourist Guide: An OR Opportunity." *OR Insight* 20: 21–27.
Verbeeck, C., P. Vansteenwegen, and E. H. Aghezzaf. 2014. "An Extension of the Arc Orienteering Problem and Its Application to Cycle Trip Planning." *Transportation Research Part E: Logistics and Transportation Review* 68: 64–78.
Wearing, S., and J. Neil. 2009. *Ecotourism: Impacts, Potentials and Possibilities?* Routledge.
Winters, M., G. Davidson, D. Kao, and K. Teschke. 2011. "Motivators and Deterrents of Bicycling: Comparing Influences on Decisions to Ride." *Transportation* 38: 153–168.
Yamanaka, H., and S. Namerikawa. 2007. "Measuring Level-of-service for Cycling of Urban Streets Using Probe Bicycle System." In Proceedings of the Eastern Asia Society for Transportation Studies The 7th International Conference of Eastern Asia Society for Transportation Studies, 2007, Eastern Asia Society for Transportation Studies. pp. 208.
Zheng, W., Z. Liao, and J. Qin. 2017. "Using a Four-step Heuristic Algorithm to Design Personalized Day Tour Route within a Tourist Attraction." *Tourism Management* 62: 335–349.
Zhu, S., and F. Zhu. 2019a. "Cycling Comfort Evaluation with Instrumented Probe Bicycle." *Transportation Research Part A: Policy and Practice* 129: 217–231.
Zhu, S., and F. Zhu. 2019b. "Multi-objective Bike-way Network Design Problem with Space–time Accessibility Constraint." *Transportation* 1–25.
Zimmermann, M., T. Mai, and E. Frejinger. 2017. "Bike Route Choice Modeling Using GPS Data without Choice Sets of Paths." *Transportation Research Part C: Emerging Technologies* 75: 183–196.